QUIZ 02 PRACTICE PROBLEMS

STAT 1110: Section S18 – Fall 2026 with Dr. Ethan P. Marzban

> **⚠ Caution**
>
> These problems are not meant to be exactly indicative of what will show up on the Quiz on Wednesday. Rather, they are designed to give you a sense of the style of multiple choice questions I typically ask. As a reminder, the quiz on Wednesday will be 15 questions.

Use the following setup for Questions 1 through 4:

Jacob wants to determine the proportion of apartments in California that are one-bedroom apartments. To that end, he goes to *Zillow* and sets his search radius to San Luis Obispo. He then assigns a unique number to each of the apartments listed, uses a random number generator to generate 10 numbers, and looks up the apartments corresponding to these 10 numbers. His 10 sampled apartments contained 4 one-bedroom apartments.

1. Describe an observational unit in the context of this dataset.

(**A**) One person

(**B**) One apartment

(**C**) One one-bedroom apartment

(**D**) Jacob

(**E**) San Luis Obispo

2. What is the sampling frame in the context of this study?

(**A**) The set of all California apartments

(**B**) The set of all California apartments listed on *Zillow*

(**C**) The set of all San Luis Obispo apartments

(**D**) The set of all San Luis Obispo apartments listed on *Zillow*

3. The value of 4/10 is important to this problem. Is it a parameter value or a statistic value?

(**A**) Parameter (**B**) Statistic

4. Suppose Jacob has carried out his sampling procedure 50 times, each time taking a sample of 10 apartments and recording the proportion that are one-bedroom. Shown below is a dotplot of his statistic values, along with the true value of the parameter (which he did not know in advance):

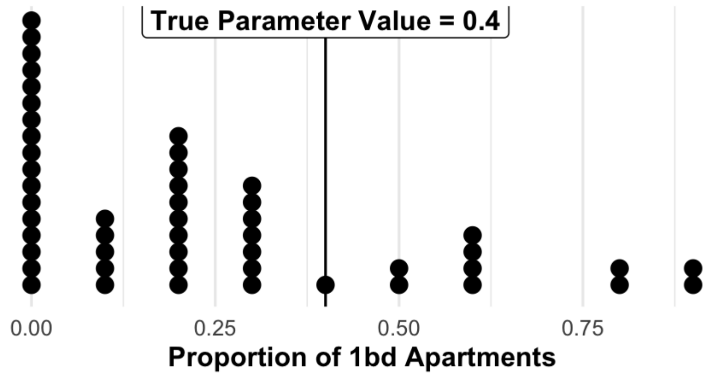

Figure 1: Dotplot of Jacob's 50 sampled proportions.

Based on this graph, do you think Jacob's sampling method is unbiased, biased with a tendency to underestimate, or biased with a tendency to overestimate?

(**A**) Unbiased

(**B**) Biased, with a tendency to underestimate

(**C**) Biased, with a tendency to overestimate

Use the following setup for Questions 5 - 9:

Pietro claims to be able to tell the number of blue M&Ms in a pack of M&Ms without ever opening the pack. Among a set of 12 packs of M&Ms, Pietro correctly identified the number of blue M&Ms in 1 of them.

5. How would you describe this study?

(**A**) Sample from a Population (**B**) Observations from a Random Process

6. What (if any) is the finite population in this study?

(**A**) The set of all packs of M&Ms in the world.

(**B**) The set of all blue M&Ms in the world.

(**C**) The set of all people named Pietro in the world.

(**D**) There is no finite population.

7. What is one observational unit?

(**A**) One pack of M&Ms

(**B**) One blue M&M

(**C**) One person

(**D**) One trial in which Pietro tries to identify the number of blue M&Ms in a pack

8. What is the sample size?

(**A**) 1/12 (**B**) 1 (**C**) 12 (**D**) Something Else

9. What is the parameter of interest?

(**A**) The proportion of a pack of M&Ms that are blue

(**B**) The proportion of all packs of M&Ms in the world that contain at least one blue M&M

(**C**) The proportion of twelve packs for which Pietro correctly identifies the number of blue M&Ms

(**D**) The long-run proportion of times Pietro correctly guesses the number of blue M&Ms in a pack

Questions 10 - 11 are unrelated:

10. The probability that a particular bus arrives at a stop late is 0.3. Which of the following best describes the interpretation of the value of 0.3?

(**A**) Out of the next 30 times the bus arrives at the bus stop, it will be late exactly 10 times.

(**B**) Out of all the times the bus arrives at the bus stop tomorrow, it will be late exactly 30% of the time.

(**C**) Among all the times the bus arrives at the bus stop over the next month, we'd expect it to be late around 30% of the time.

11. Marissa takes several samples of 12 cats and records the standard deviation of weights in each sample. She then uses her sample standard deviation values to create the following dotplot:

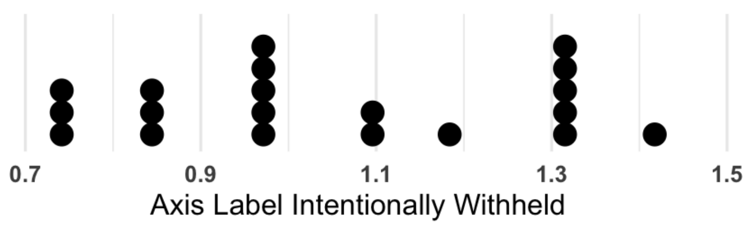

Figure 2: Dotplot of Marissa's Data.

What does one dot on this dotplot represent?

(**A**) The weight of one cat

(**B**) One cat

(**C**) The standard deviation of a sample of 12 cat weights

(**D**) The standard deviation among all cats in Marissa's sampling frame

(**E**) Something else